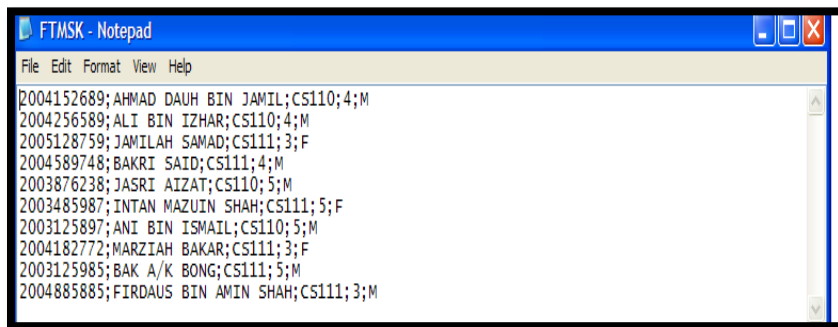


CSC186 – Object Oriented Programming
Academic Session October 2025 – February 2026
Lab Assignment 4 – File Input and Output

Course Outcomes (CO)	LO1	LO2	LO3
CO1			
CO2	√	√	√
CO3			

- 3.1 Given the input file of `FTMSK.txt` that includes the following attributes such as MatricNumber, StudentName, Program, Part and Gender (either M (Male) or F (Female)).

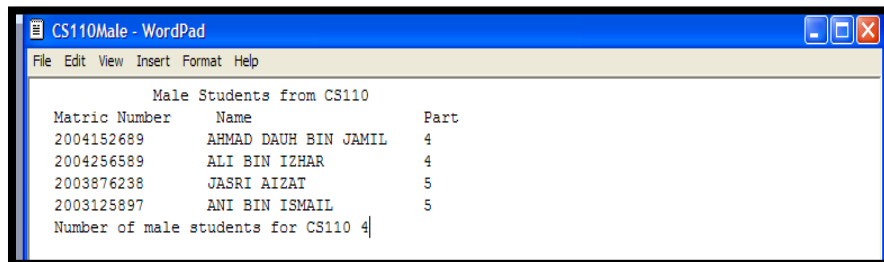
The following input text file includes all records of FTMSK students.



```
2004152689;AHMAD DAUH BIN JAMIL;CS110;4;M
2004256589;ALI BIN IZHAR;CS110;4;M
2005128759;JAMILAH SAMAD;CS111;3;F
2004589748;BAKRI SAID;CS111;4;M
2003876238;JASRI AIZAT;CS110;5;M
2003485987;INTAN MAZUIN SHAH;CS111;5;F
2003125897;ANI BIN ISMAIL;CS110;5;M
2004182772;MARZIAH BAKAR;CS111;3;F
2003125985;BAK A/K BONG;CS111;5;M
2004885885;FIRDAUS BIN AMIN SHAH;CS111;3;M
```

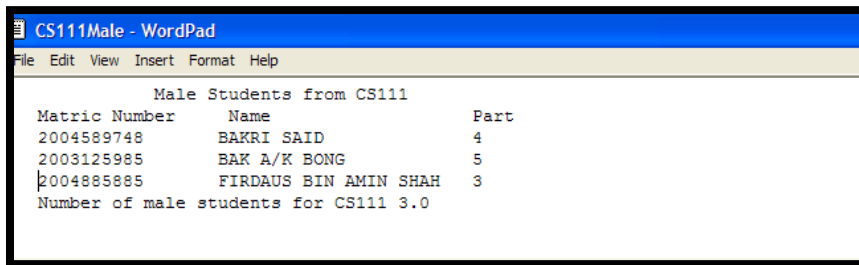
Write a program that reads each record from `FTMSK.txt` and filter the records into two output text files called `CS110Male.txt` and `CS111Male.txt`. The first output text file will store all male students from CS110, while the second output file will store all male students from CS111. Calculate and display the total records for each output text file.

Examples of output text file for `CS110Male.txt`.



```
Male Students from CS110
Matric Number    Name                Part
2004152689      AHMAD DAUH BIN JAMIL  4
2004256589      ALI BIN IZHAR        4
2003876238      JASRI AIZAT          5
2003125897      ANI BIN ISMAIL        5
Number of male students for CS110 4
```

Examples of output text file for CS111Male.txt.

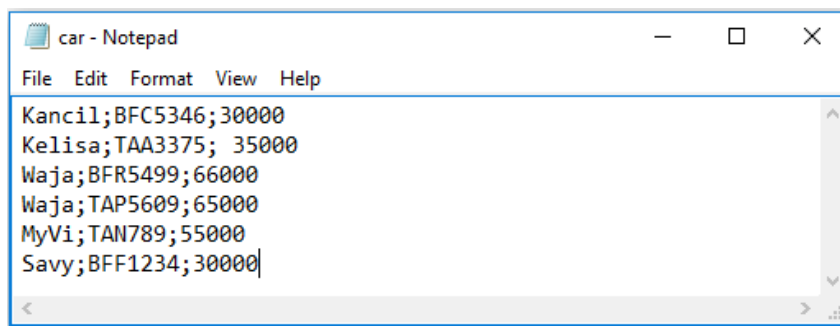


```
Male Students from CS111
Matric Number   Name           Part
2004589748      BAKRI SAID     4
2003125985      BAK A/K BONG   5
2004885885      FIRDAUS BIN AMIN SHAH 3
Number of male students for CS111 3.0
```

Use exception-handling operations to deal with the input-output errors.

- 3.2 H&M Car Motors Bhd is in the process of updating the record of cars information. You are given a file named `car.txt` that contains the following details:

- car type
- car plate number
- car price



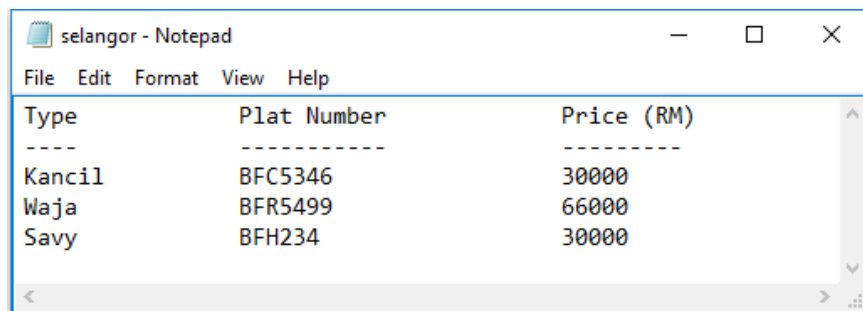
```
Kancil;BFC5346;30000
Kelisa;TAA3375; 35000
Waja;BFR5499;66000
Waja;TAP5609;65000
MyVi;TAN789;55000
Savy;BFF1234;30000
```

A sample of `car.txt` file.

Write a complete program to perform the following:

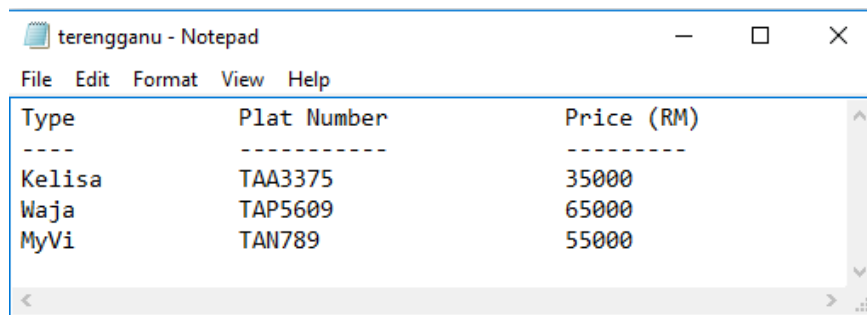
- Design a complete class named `Vehicle` consist of the following method:
 - Constructor
 - Setter/mutator
 - Getter/Assessor
- Write Java Application program to perform the following tasks:
 - Create an array of object for `Vehicle` named `arrCar`.
 - Read all records from the file named `car.txt` and store each object onto array of object named `arrCar`

- iii) Write the following information into a file named `selangor.txt`:
- Information for all cars that are manufactured in Selangor, where the plate numbers start with the letter 'B'.
 - Calculate the total price for all cars that are manufactured in Selangor.
- iv) Write the following information into a file named `terengganu.txt`:
- Store the information for all cars that are manufactured in Terengganu, where the plate numbers start with the letter 'T'.
 - Calculate the total price for all cars that are manufactured in Terengganu.
- v) Use exception to handle possible file errors and wrong data type errors.



Type	Plat Number	Price (RM)
Kancil	BFC5346	30000
Waja	BFR5499	66000
Savy	BFH234	30000

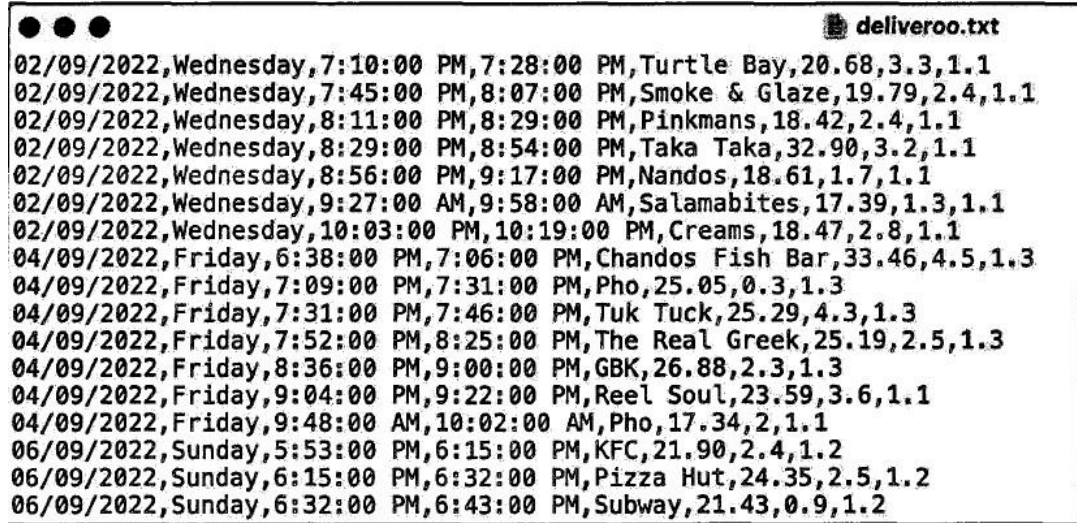
Sample of `selangor.txt`



Type	Plat Number	Price (RM)
Kelisa	TAA3375	35000
Waja	TAP5609	65000
MyVi	TAN789	55000

Sample of `terengganu.txt`

- 3.3 You are given a text file named `deliveroo.txt`, which contains a list of food delivery orders for a food delivery company. The data consists of the delivery date (DDMMYYYY), day, start time, end time, restaurant name, total price, delivery distance in kilometres, and price multiplier.



```
02/09/2022,Wednesday,7:10:00 PM,7:28:00 PM,Turtle Bay,20.68,3.3,1.1
02/09/2022,Wednesday,7:45:00 PM,8:07:00 PM,Smoke & Glaze,19.79,2.4,1.1
02/09/2022,Wednesday,8:11:00 PM,8:29:00 PM,Pinkmans,18.42,2.4,1.1
02/09/2022,Wednesday,8:29:00 PM,8:54:00 PM,Taka Taka,32.90,3.2,1.1
02/09/2022,Wednesday,8:56:00 PM,9:17:00 PM,Nandos,18.61,1.7,1.1
02/09/2022,Wednesday,9:27:00 AM,9:58:00 AM,Salamabites,17.39,1.3,1.1
02/09/2022,Wednesday,10:03:00 PM,10:19:00 PM,Creams,18.47,2.8,1.1
04/09/2022,Friday,6:38:00 PM,7:06:00 PM,Chandos Fish Bar,33.46,4.5,1.3
04/09/2022,Friday,7:09:00 PM,7:31:00 PM,Pho,25.05,0.3,1.3
04/09/2022,Friday,7:31:00 PM,7:46:00 PM,Tuk Tuck,25.29,4.3,1.3
04/09/2022,Friday,7:52:00 PM,8:25:00 PM,The Real Greek,25.19,2.5,1.3
04/09/2022,Friday,8:36:00 PM,9:00:00 PM,GBK,26.88,2.3,1.3
04/09/2022,Friday,9:04:00 PM,9:22:00 PM,Reel Soul,23.59,3.6,1.1
04/09/2022,Friday,9:48:00 AM,10:02:00 AM,Pho,17.34,2,1.1
06/09/2022,Sunday,5:53:00 PM,6:15:00 PM,KFC,21.90,2.4,1.2
06/09/2022,Sunday,6:15:00 PM,6:32:00 PM,Pizza Hut,24.35,2.5,1.2
06/09/2022,Sunday,6:32:00 PM,6:43:00 PM,Subway,21.43,0.9,1.2
```

Figure 1: `deliveroo.txt`

Write a complete Java program using File Input and Output operation to perform the following tasks:

- Read all records from the text file named `deliveroo.txt` in Figure 1.
- Generate an output file named `earnings.txt` which contains the profit and summary of the orders. The profit for each delivery is calculated using the following formula:

$$\text{profit} = (\text{price multiplier} \times \text{final price}) - \text{final price}$$

The final price that the buyer must pay is the sum of the order's total price multiplied by the price multiplier. The price multiplier was set based on the type of customers and the coupon they used.

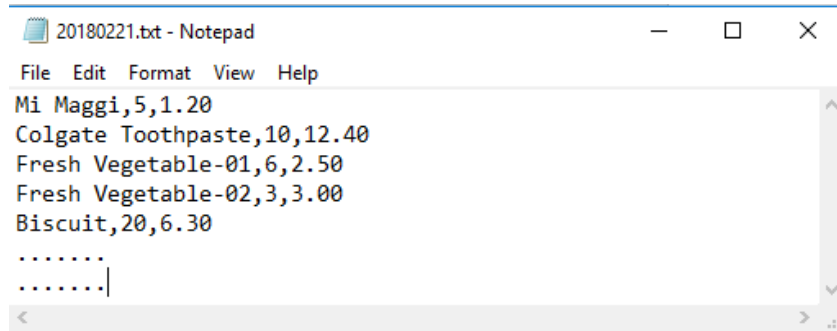
A sample of `earnings.txt` is shown in Figure 2.

Profit earned for each delivery	
Date(Day)	Profit
=====	
02/09/2022(Wed)	RM2.07
02/09/2022(Wed)	RM1.98
02/09/2022(Wed)	RM1.84
02/09/2022(Wed)	RM3.29
02/09/2022(Wed)	RM1.86
02/09/2022(Wed)	RM1.74
02/09/2022(Wed)	RM1.85
04/09/2022(Fri)	RM10.04
04/09/2022(Fri)	RM7.52
04/09/2022(Fri)	RM7.59
04/09/2022(Fri)	RM7.56
04/09/2022(Fri)	RM8.06
04/09/2022(Fri)	RM2.36
04/09/2022(Fri)	RM1.73
06/09/2022(Sun)	RM4.38
06/09/2022(Sun)	RM4.87
06/09/2022(Sun)	RM4.29
=====	
Average profit for each order: RM4.94	

Figure 2: earnings. txt

- c) Count the number of deliveries with start time at 9 PM or later and display it on the console.
- d) Close all file streams in use.
- e) Apply exception handling mechanism in the program.

- 3.4 Ahmad Jimat Supermarket is a convenient store that sells varieties of goods to its customers. All sales are recorded in text files. The text files are named according to the date of the sales. The stored data contains item name, quantity and unit price. The file is named "20180221.txt" records all the sales on 21st February 2018 as follows:



A sample of 20180221.txt

Write a complete Java program using file I/O to perform the following tasks:

- a) Design a complete class named `Supermarket` consist of the following method:
 - i) Constructor
 - ii) Setter/mutator
 - iii) Getter/Assessor
- b) Write Java Application program to perform the following tasks:
 - i) Create an array of object for named `arrStore`.
 - ii) Read all records from the file named `20180221.txt` and store each object onto array of object named `arrStore`
 - iii) Write the following into a file named `reportSale.txt`:
 - Calculate and display the total sale on 21st February 2018 by summing up the subtotal of all the records in the text file for each record. For example, the subtotal of the first record is shown below:
$$\text{Sub total} = 5 * 1.20 = 6.00$$
 - The expected output of the program is shown below:

```
reportSale.txt - Notepad
File Edit Format View Help
Item          Quantity    Price(RM)    Sub Total(RM)
Mi Maggi      5           1.20         6.00
Colgate Toothpaste 10        12.40       120.00
Fresh Vegetable-01 6          2.50        15.00
.....      ..          ...          ....
.....      ..          ...          ....

TOTAL SALE for date: 21st, February 2018 is: RM xxx.xx.
```

A sample of reportSale.txt

- iv) Use exception handling operations to deal with possible file input/output error.

- 3.5 The Academic Affairs Department of Cemerlang College provides photocopy services for the lecturers. To request the service, the lecturers will need to provide their name, staff id, faculty code (FHTM-Faculty of Hotel and Tourism, FSSR-Faculty of Sports Science and Recreation), type of assessment (quiz, test, examination), number of pages for master copy and number of copies required. All the data will be stored in a file named `photocopyInfo.txt`. The following class `Photocopy` defines all the details required for the photocopy service.

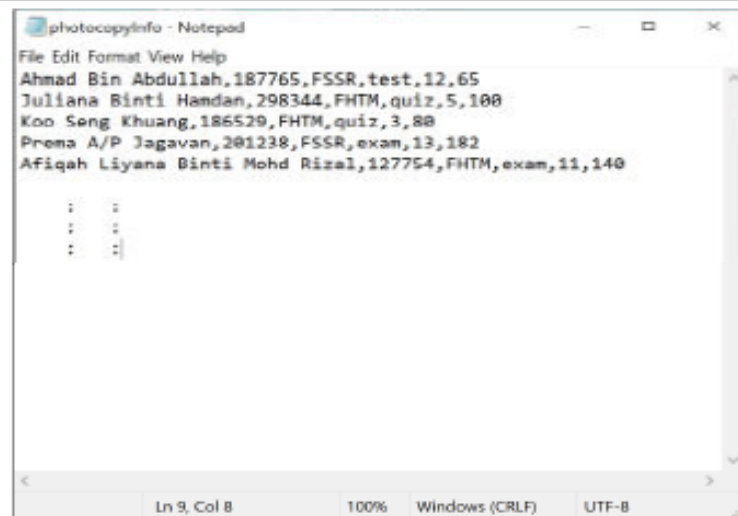
Class Name : `Photocopy`

Attributes :

```
String lectName;  
String staffID;  
String facCode;    //faculty code either FHTM(Faculty of Hotel and  
                   //Tourism) or FSSR(Faculty of Sports Science and  
                   //Recreation).  
  
String assType;    //either quiz or test or exam.  
int masterNum;    //number of pages of master copy.  
int copiesNum;    //number of copies.
```

Behavior :

```
public Photocopy() {...} //default constructor  
public void setPhotocopy(...) {...} //setter method  
  
//retriever methods  
public String getLectName() {...}  
public String getStaffID() {...}  
public String getFacCode() {...}  
public String getAssType() {...}  
public int getMasterNum() {...}  
public int getCopiesNum() {...}  
  
public String toString() {...} //display method
```

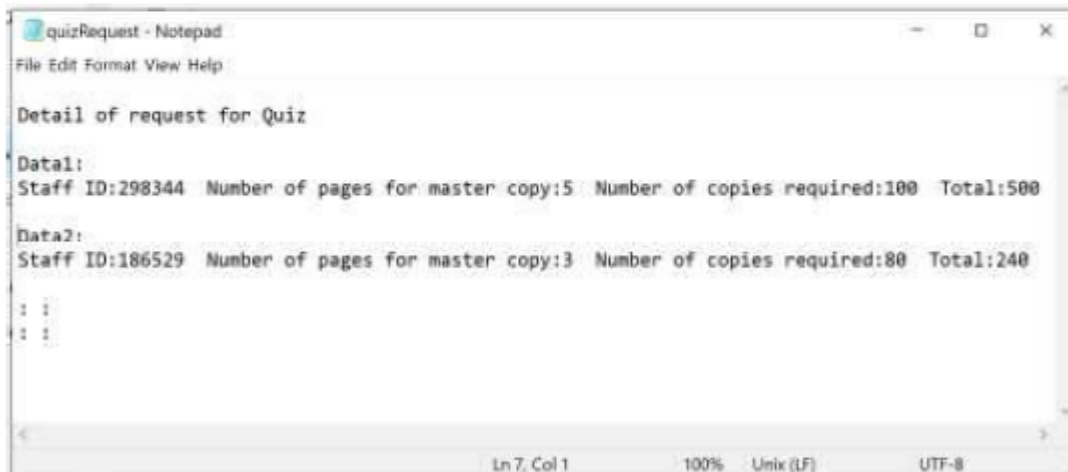


```
photocopyInfo - Notepad  
File Edit Format View Help  
Ahmad Bin Abdullah,187765,FSSR,test,12,65  
Juliana Binti Hamdan,298344,FHTM,quiz,5,100  
Koo Seng Khuang,186529,FHTM,quiz,3,80  
Prenna A/P Jagavan,201238,FSSR,exam,13,182  
Afiqah Liyana Binti Mohd Rizal,127754,FHTM,exam,11,140  
  
:  
:  
:  
:  
  
Ln 9, Col 8    100%    Windows (CRLF)    UTF-8
```

`photocopyInfo.txt`

Write Java program segments to do the following:

- a) Open an input file named `photocopyInfo.txt`, read all the data and store them into an array of object, `copiesArr` of type `Photocopy`. This array can store up to 200 records, but the actual storage used is based on number of records in the input file.
(5.5 marks)
- b) Display all information of photocopy services requested by the lecturers of FHTM.
(1.5 marks)
- c) Find all photocopy requests for quiz and copy the staff id, number of pages for master copy, number of copies required and total number of copies (number of pages for master copy multiply with number of copies required) into an output file named `quizRequest.txt` as depicted in the following diagram.



```
quizRequest.txt
```

(4 marks)

- d) Each photocopy uses A4 papers and two-sided copying mode. Each ream has 500 pieces of A4 paper. Calculate and display number of reams needed to meet all the photocopy requests.

(4 marks)